Saikiran Juttu

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Education

Northeastern University, Boston, MA Dec 2025
Master of Science in Robotics — GPA: 3.99/4.0
Relevant Coursework: Pattern Recognition and Computer Vision, Control Systems, Robotics Sensing and Navigation, Mobile Robotics, Autonomous Field Robotics, Robot Mechanics and Control, Reinforcement Learning

Dr. B.R. Ambedkar National Institute of Technology, Jalandhar, India May 2020
Bachelor of Technology in Mechanical Engineering
Relevant Coursework: Mechatronics, CAD, Material Science, Industrial Automation, Machine Design, Operations Research, Electronics

Skills

- **Languages**: C++, Python, C
- **Tools**: MATLAB, Simulink, ROS/ROS2, Linux, Gazebo, Docker, Git, SolidWorks, CATIA, CAD, Gymnasium, ArduPilot, MuJoCo
- **Libraries & Frameworks**: OpenCV, NumPy, PyTorch, PCL, TensorFlow, AWS
- **Hardware Proficiencies**: 3D Printing, PCB Designing, Hardware Testing, Arduino Uno, Raspberry Pi, Jetson, PLC Programming

Experience

Graduate Research Assistant, Multi-Agent Robotics and Autonomy Laboratory — Boston, MA Apr 2024 – Present
- Designing a heterogeneous UAV–UGV cooperative exploration framework in **Gazebo (ROS/ROS2, C++)**, integrating exploration algorithms, path-planning strategies, sensor data fusion, and tracking controllers for multi-robot coordination
- Developing a unified incremental PRM-based planner in **C++/Python**, improving exploration efficiency and scalability
- Designing and testing custom FPV drone hardware, tuning flight controllers and collecting flight data to improve performance in GPS-denied environments
- Implemented vision-based object detection, IK solver, and motion planning in **MATLAB**, boosting robotic arm precision by 30%
- Modeled a high-fidelity **Simscape/MATLAB** manipulator simulation and a physical robotic arm testbed, enabling remote-access experimentation and reducing hardware deployment time by 25%

Robotics Engineer Co-op, Northeastern University — Boston, MA Jan 2025 – Jun 2025
- Integrated 3D LiDAR and Intel RealSense sensors with the Scout Mini Rover via CAN protocol in **ROS 2**, deploying on embedded platforms (**Jetson AGX Orin**) to enable synchronized perception, sensor fusion, and reliable hardware–software communication
- Implemented **ROS 2**-based SLAM solutions (RTAB-Map, LIO-SAM, VIO, SLAM Toolbox) in **C++/Python**, leveraging **Nav2** to improve localization accuracy and enable robust autonomous navigation in GPS-denied environments
- Developed waypoint navigation and motion planning pipelines for UAVs (Crazyflie, DJI Tello) using the **OptiTrack motion capture system**, automated takeoff, and precise landing systems
- Exploring integration of **Vision-Language Models (VLMs)** with onboard perception to enhance semantic understanding and indoor navigation, targeting reduced localization errors and improved autonomy

Design Engineer, Technip Energies — Noida, India Feb 2021 – Jul 2023
- Designed and optimized large-scale piping systems using **Smart3D, Navisworks, SolidWorks, CATIA**, gaining expertise in CAD modeling, system integration, and safety-compliant mechanical design
- Developed a predictive Load Monitoring System with **Python/RPA**, reducing downtime by 20% and improving reliability
- Spearheaded RPA-based workflow automation, streamlining business processes and cutting operational man-hours by 4%
- Gained expertise in large-scale system engineering and validation through **EPC projects**, building a foundation for developing robust, safety-critical robotic systems

Computer Vision Intern, Robic Rufarm India Pvt Ltd — Hyderabad, India Jun 2019 – Aug 2019
- Built a real-time defect detection and classification system in **Python/OpenCV**, achieving 90% accuracy and significantly enhancing production throughput
- Optimized live video analysis under variable lighting using **image processing techniques**, boosting detection accuracy to over 90% and reducing manual inspection time by 50%

Projects

Photo Mosaicking of Low-Contrast Underwater Images — OpenCV, Python, GTSAM
- Developed a robust image registration pipeline using SIFT, RANSAC, and Levenberg-Marquardt algorithms, enabling accurate feature matching and homography estimation for low-contrast underwater datasets
- Built and optimized factor graphs in GTSAM to detect loop closures, enhance odometry accuracy, and refine covariance estimates.

Sparse 3D Reconstruction and Bundle Adjustment — Python, OpenCV, GTSAM
- Built a custom pipeline consisting of algorithms to compute Fundamental and Essential Matrices, triangulate 3D points using Epipolar Geometry, solve Perspective-n-Point (PnP) problems, and perform Bundle Adjustment for sparse reconstruction
- Performed bundle adjustment on 3D points and camera positions using GTSAM, improving reconstruction accuracy by 20%

Automated Insect Leg Labeling — C++, OpenCV, Python, PyTorch
- Developed an automated labeling pipeline for insect legs using image processing, feature detection, and clustering within DeepLabCut
- Reduced labeling time by more than 70% while maintaining high precision and accuracy, accelerating data preparation for research

Autonomous Race Driving with Action Mapping RL — Python, PyTorch, NumPy, Matplotlib
- Implemented reinforcement learning algorithms (**DDPG, TD3, SAC**) with an **Action Mapping** mechanism in a custom racing simulator, incorporating a **friction-aware reward function** to enforce vehicle dynamics and tire-road constraints
- Validated TD3 against baseline DDPG, achieving up to **11% faster lap times** and smoother trajectories, demonstrating superior policy stability and constraint adherence